

Action understanding with presentational goals

Anonymous CogSci submission

Abstract

Humans balance many goals while navigating social interactions. We are motivated not only by what we want for ourselves, and what we want for others, but also by what we want others to think about us. Inferences about agents' personal and social goals have been formally captured using Bayesian inverse planning models. However, less is known about the mechanisms underlying our ability to infer *presentational goals*: an agent's desires over how another agent sees them. We introduce a novel paradigm which allows us to test joint inferences about social and presentational goals. We propose an extension to inverse planning where agents derive additional utility from shifting others' beliefs about their social goals. Because the model places no constraint on the valence of presentational targets, it naturally captures cases where agents wish to appear prosocial or adversarial. Across a variety of scenarios, participants make systematic joint inferences about social goals, presentational goals, and presentational targets. Our computational model captures participants' inferences better than feature-based alternatives.

Keywords: theory of mind; action understanding; social cognition; Bayesian inference; presentational goals

Introduction

Imagine yourself in a park. You glance at a play area and see two children, Alice and Bob. Alice teases Bob and takes his toy despite his protestations. You drift off in a book and lift your head a moment later as Charlie, Alice and Bob's parent, arrives and waves at you. As you turn your gaze back towards Alice and Bob, you see her helping Bob with his toy, carefully caring for him. You gently smile at this sight, noticing Alice's efforts to please Charlie, without thinking much of it. Yet, the inference you just drew is anything but trivial. It rests on rich assumptions about what motivates us in social relationships, such as presentation of self and reputation.

Humans are able to draw rich inferences from sparse data when interpreting the behaviour of others. Previous research suggests that, to achieve this, people invert a generative model of what a rational agent would do to determine others' goals and preferences (Baker et al., 2009, 2017). This perspective offers a powerful explanation for how we draw inferences from little evidence by proposing that we leverage a robust yet naive generative model of what motivates people: they act to maximise their utility (Jara-Ettinger et al., 2016).

Early work focused on agents with solely self-interested goals, such as reaching a desired location or object. More recent research has incorporated social goals that involve adopt-

ing another agent's utility into one's own (Baker et al., 2008; Powell, 2022; Ullman et al., 2009). Inverse planning models account well for people's intuitive inferences about these kinds of social goals (Netanyahu et al., 2021; Shu et al., 2020; Tejwani et al., 2022). Not only are people apt at making inferences about whether an agent wants to harm or help another, but they can also use them as input to richer moral judgments such as responsibility and blame (Wu et al., 2023).

Yet, navigating social interactions and maintaining interpersonal relationships can often involve even higher order goals over the mental states of others. Impression management involves holding preferences over others' beliefs of ourselves, e.g. we may want others to think we are prosocial, inequity averse, or competent (Goffman, 1959; Kleiman-Weiner et al., 2017; Zhao et al., 2021). Children from a young age hold preferences about what adults think of their competence and attempt targeted demonstrations of ability to move the beliefs of adults in desirable states (Gweon, 2021). Crucially, people happily trade these objectives off with material outcomes by signalling an intention to collaborate by forgoing immediate reward (Battigalli & Dufwenberg, 2022; Roberts, 2020). Computational accounts of such higher order goals have had to take seriously the idea that targeted changes in the belief states of others can constitute meaningful sources of reward for people, thus naturally drawing conceptual links to rational theories of communication (Chandra et al., 2024; Degen, 2023; Yoon et al., 2020).

The present work focuses on higher order goals defined over what other people believe about our social goals. Recent research has proposed that presentational goals, that is, attempting to control the impression we give to others, may be formulated as deriving utility from maintaining the beliefs others have of how much we weigh different rewards when considering the overall utility of taking an action (Houlihan et al., 2023; Kleiman-Weiner et al., 2017). This view proved useful in showing that people are sensitive to how ready others are to change their mind when considering presentational interventions. This leads them to act more selfishly when they feel others are convinced of their good intentions (Btesh et al., 2025).

Here, we study inferences about agents' social and presentational goals in a novel experimental paradigm that allows us to pit them directly against each other, and also against personal goals. These three types of goals may not always be aligned, which can produce counter-intuitive yet informative

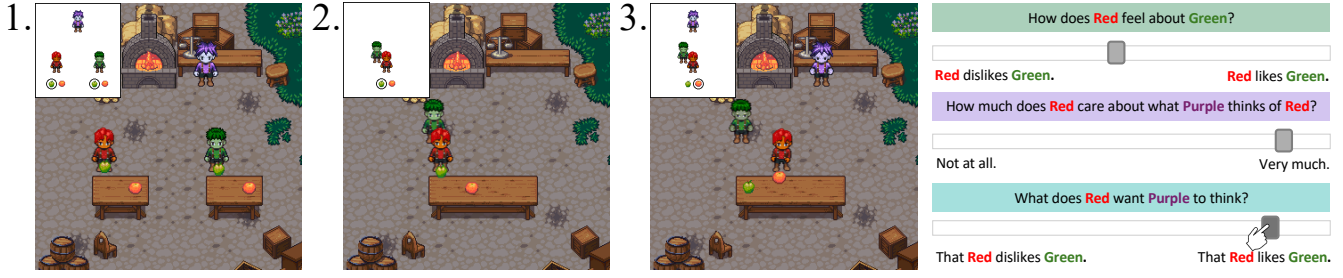


Figure 1: **An example story** (story A. in Figure 2) as it was presented to participants with the questions that participants were asked to answer. We expected participants to interpret Red as trying to convince Purple that they like Green. 1. Red and Green both prefer apples. Purple sees them choose and now knows what they each prefer. Red also knows what Green prefers. 2. Red and Green wait in line. Red goes first and takes the apple. Purple is absent. 3. Red and Green wait in line. Red goes first and takes the orange. Purple sees them choose.

behaviour. Nevertheless, we find that people are able to make systematic joint inferences from just a few observations, and we develop a computational model that extends the naive utility calculus and accurately captures patterns of participants’ judgments.

Experimental paradigm

We introduce a new theory of mind task to systematically assess people’s intuitive theories about how people trade off personal, social, and presentational goals. Crucially, the paradigm allows for situations in which the different goals can come apart, for example, where Alice does not like Bob but wants Charlie to think otherwise.

Consider the scenario in Figure 1. In the first scene, all characters are present and Red and Green each take an apple from their table, indicating that they like apples more than oranges. Purple sees them both choose and now knows their preference. Crucially, Red also knows that Purple knows Red and Green’s preference. In the second scene, Red and Green stand in line. Purple is absent and Red grabs the apple, knowing that that’s also Green’s preferred fruit. Finally, in the last scene, Purple is present again but this time, Red chooses the orange, leaving the apple for Green.

Notice how this story shares the same features as the two children in the park: Red is nice when a third party is present, but not so much otherwise. In our task, participants see stories involving a Red, a Green, and a Purple character. Each story consists of three scenes, initially shown as short clips of the interaction, then summarised in images. The first scene shows Red and Green independently choosing between an apple and an orange, in Purple’s and each others’ presence. This scene signals which fruit each character prefers. The second scene shows Red and Green waiting in line to pick a fruit. Red always picks first and chooses between either an apple and an orange or between two apples and an orange, leaving Green with the remaining fruit(s) (Green’s choice afterwards is not shown). This scene thus provides information about Red’s social goal, as Purple is not able to see the scene. The third scene shares the structure of the second, but adds the presence of Purple, thus providing an opportunity to observe whether Red’s behaviour changes in the presence of Purple. At the end of all three scenes, we ask participants to report judgements

about Red’s personal, social, and presentational goals (see Figure 1, right).

We hypothesise that participants will use inverse planning to decode the behaviour of Red in each story. We present below the model which we expect to capture people’s intuitive theory about Red’ motivations.

Computational model

Drawing from previous literature on inverse planning applied to presentation of self, we hypothesise that when trying to decode the intentions of actors, people will assume that the actions of sophisticated Red (with presentational goals) are driven by the following utility:

$$U_{\text{sophisticated}}(f) = V_{\text{red}}(f, \alpha_{\text{red}}) - \beta \cdot V_{\text{green}}(f, \alpha_{\text{green}}) \cdot L_{\text{green}}(k, n_f) - \delta \cdot D_{KL}[p_{\text{purple}}(\beta|f) || \rho(\beta_{\text{goal}})] \quad (1)$$

where $f \in \{\text{apple}, \text{orange}\}$, $\beta \in [-1, 1]$ controls how prosocial Red is. The negative sign in front of β represents the fact that the choice f for Red is actually loss for Green if Green prefers f as well. Note that $\rho(\beta_{\text{goal}})$ is not a prior but the target distribution over what Red would want Purple to believe their social goal for Green is, i.e. $p(\beta|f)$, where $\beta = -1$ means they dislike Green and $\beta = 1$ means that they like Green. $\delta \in [0, 1]$ controls how concerned Red is that Purple perceives their prosociality to be congruent with their goal $\rho(\beta_{\text{goal}})$.

Notice the negative sign before δ : the agent is to minimise the KL-divergence between the posterior distribution others would have if they did action a and their target distribution. Here, the desire to be seen as prosocial is instantiated via making people believe that we promote others’ utility in our own. This model implies that we would be motivated to act prosocially in situations where we believe our actions would change others’ belief about our prosociality, not otherwise.

Finally, we highlight the conceptual difference between δ , which controls whether Red has a presentational goal, i.e. whether they care about what others think of them, and β_{goal} , which determines the presentational *target*, that is, what Red would want others to think about their prosociality. Note here that our model does not assume that agents want to be seen as prosocial. In fact, it lets Red have any presentational goal over β , while δ controls how important the goal is to Red.

This model is therefore not restricted to modeling positively valenced presentational goals and admits in principle that Red could want to hide that they care about Green ($\beta_{\text{goal}} = 0$), or make Purple think that they do not like Green ($\beta_{\text{goal}} = -1$).

The utility that each character derives from fruit f is:

$$V_{\text{red}}(f, \alpha_{\text{red}}) = \begin{cases} \alpha_{\text{red}} & \text{if } f = 1 \\ 1 - \alpha_{\text{red}} & \text{otherwise} \end{cases} \quad (2)$$

where $\alpha_{\text{red}} \in [0, 1]$ represents how much Red prefers apples. It allows to map discrete fruit choices and a continuous preference to a continuous utility. Also, $L_{\text{green}}(k, n_f) = \min(1, 1 + k - n_f)$ and computes the loss that Green would incur if Red picks a fruit f . n_f is the number of available fruit f . This lets Green incur no loss should two apples be present and Red picks one of them if $k = 1$ and a loss of one otherwise.

Let us now consider the beliefs at play here. For simplicity, we assume that $p(\beta, \delta, \beta_{\text{goal}})$ is uniform, that is, the observer does not assume anything about Red’s prosociality and presentational goals. Beyond the three parameters they are trying to estimate, the observer still has uncertainty about Red. First, they represent how tied Red is to their preference α_{red} , given their choice in the first scene. To model this, the observer has a belief $p(\alpha_{\text{red}}|f_{\text{red}}) = \text{Beta}(f_{\text{red}} \cdot \tau_{\text{obs}}, (1 - f_{\text{red}}) \cdot \tau_{\text{obs}})$, where f_{red} is a binary variable indicating Red’s chosen fruit in the first scene (0 for apple and 1 for orange) and τ_{obs} is a precision parameter. Then $f_{\text{red}} \cdot \tau_{\text{obs}}$ and $(1 - f_{\text{red}}) \cdot \tau_{\text{obs}}$ act as pseudo-counts for the Beta distribution.

From this point, the observer has to model Red’s own uncertainty about the situation. First, Red can be uncertain how tied Green is to their own preference α_{green} , given Green’s choice in the first scene. This is represented as $p(\alpha_{\text{green}}|f_{\text{green}}) = \text{Beta}(f_{\text{green}} \cdot \tau, (1 - f_{\text{green}}) \cdot \tau)$, where τ is a precision parameter, distinct from τ_{obs} . Then, Red has uncertainty about how many apples Green would prefer to have, i.e. k , that is how much subjective loss would Green incur in stories where 2 apples and 1 orange are available should Red pick one of the apples. Here, we simply assume a binary distribution over 1 or 2 apples parameterised by probability θ .

However, note that Equation 1 abstracts a key point of complexity. Specifically, evaluating the presentational goal requires that Red thinks that Purple holds a model of Red’s own utility. Btsh et al. (2025) assumed that from Red’s perspective, Purple sees Red as acting according to:

$$U_{\text{simple}}(f) = V_{\text{red}}(f, \alpha_{\text{red}}) - \beta \cdot V_{\text{green}}(f, \alpha_{\text{green}}, k) \cdot L_{\text{green}}(k, n_f) \quad (3)$$

which is simply Equation 1 without the presentational goal. This overall, yields four new beliefs: $p_{\text{purple}}(\beta)$, $p_{\text{purple}}(\alpha_{\text{red}})$, $p_{\text{purple}}(\alpha_{\text{green}})$ and $p_{\text{purple}}(k)$. Where the underscore indicates that these are Purple’s beliefs about Red, keeping in mind that we are within the perspective of Red.

For simplicity, we assume a uniform prior for $p_{\text{purple}}(\beta)$, that is Purple is unsure about Red’s prosociality. However, as Red knows that Purple has seen Red and Green’s choices in the first scene, Purple knows what their preferences are. Therefore, we set $p_{\text{purple}}(\alpha_{\text{red}}) = \text{Beta}(f_{\text{red}} \cdot \tau, (1 - f_{\text{red}}) \cdot \tau)$

and $p_{\text{purple}}(\alpha_{\text{green}}) = \text{Beta}(f_{\text{green}} \cdot \tau, (1 - f_{\text{green}}) \cdot \tau)$, where τ is assumed to be the same as previously introduced, representing Red’s overall uncertainty about the situation. This then allows Red to use Bayes’ rule to compute the posterior distribution $p_{\text{purple}}(\beta|f)$ that Purple would have if Red picked f .

The last piece to compute the presentational goal is the presentational target $\rho(\beta_{\text{goal}}) = \mathcal{N}(\beta_{\text{goal}}, \sigma_{\text{goal}}^2)$ where σ_{goal} represents how tight Red’s presentational target is around β_{goal} , that is, how sure do they want Purple to think that $\beta = \beta_{\text{goal}}$.

Given all those components, the model conditions on the number of fruits and the preferences f_{red} and f_{green} and on the Red’s subsequent choices f_1 and f_2 and provides a joint posterior distribution over $\beta, \delta, \beta_{\text{goal}}$ given a story.

Model variants

Our prediction is that participants answer each question by assuming that Red chooses fruits f_1 and f_2 according to our proposed generative model and invert it to infer a joint posterior over each parameter: $p(\beta, \delta, \beta_{\text{goal}}|f_1, f_2)$.

Marginals from the joint posterior The first natural way through which they could then choose their response is to sample from each parameter’s respective marginal distribution. This constitutes our first model. This model has 5 parameters. First, the precision of the observer τ_{obs} , Red’s precisions τ , the likelihood that Green eats two rather than one apple θ , how precise the presentational target $\rho(\beta_{\text{goal}})$ is σ_{goal} and finally an inverse temperature parameter T^{-1} for a softmax through which the posterior marginal distributions are passed and which represents the rationality of participants’ responses.

Estimating δ by conditioning on $\beta_{\text{goal}} = 1$ While the joint model captured participants’ estimates for β and β_{goal} quite well, it totally failed for δ and the posterior distributions tended to be close to uniform due to how little data is available for inference. We thus propose that participants rely on stronger assumptions to make their judgments. Here, we simply propose that instead of marginalising out δ , participants instead condition on a $\beta_{\text{goal}} = 1$, that is they answer: if Red wanted to be well seen, did they act in a way which is consistent with wanting Purple to believe it? This second model shares all parameters with the first one.

Heuristic for δ Lastly, we propose an even stronger assumption. Given that estimating δ seemed complex, we thought that some participants may rely on a heuristic: if Red’s behaviour changes in the presence of Purple, then $\delta = 1$ otherwise $\delta = 0$. To represent this we add two parameters, a mixing weight α controlling the proportion of people who answer the question in this way and the noise in those participants responses $\sigma_{\text{heuristic}}$, yielding a total of 7 parameters for this last model.

Feature-based model To test whether participants were answering each question using the features of each story rather than doing inverse planning, we fit three linear mixed effects

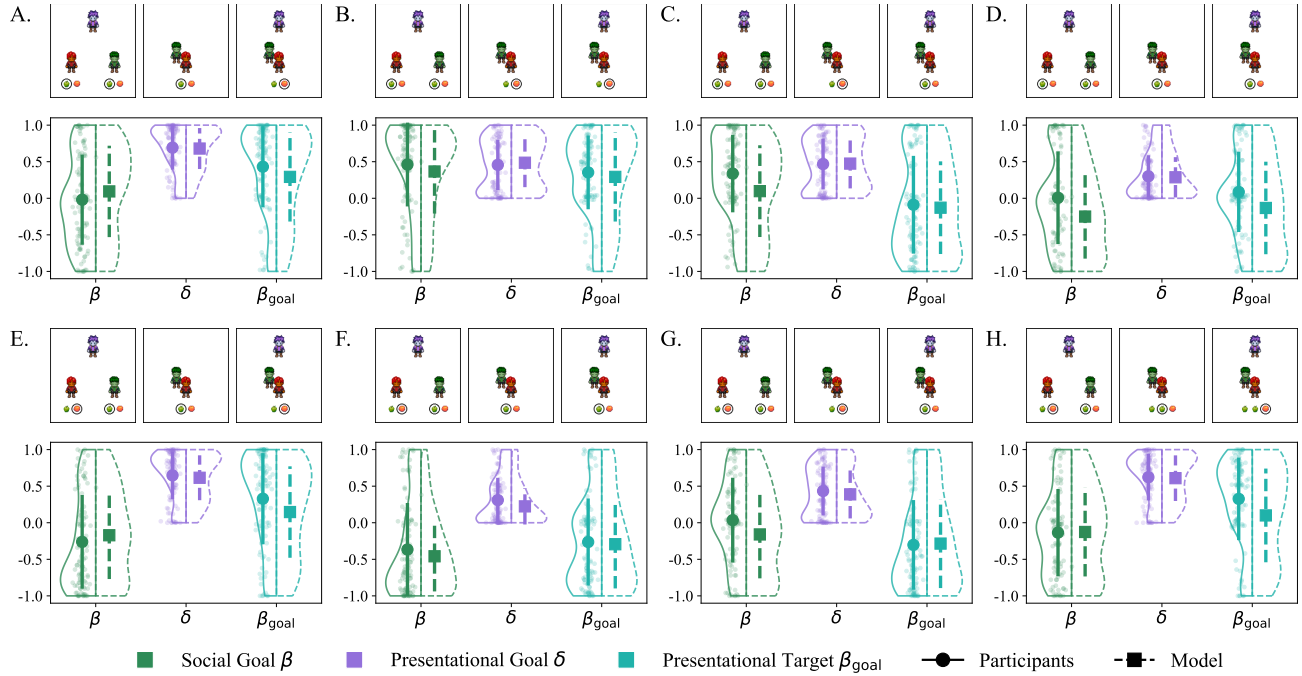


Figure 2: **Distributions of participants and model responses for selected stories.** The three images at the top of each plot depicts each scene of the story being modelled. Violin plots represent the posterior marginal distributions for each judgments, points represent means and error bars are standard deviations. Full lines are participants and dotted lines model predictions for the model which conditions on β_{goal} and uses the heuristic when Red changes their behaviour in the presence of Purple.

models predicting β , δ and β_{goal} from story data. The variables were binary coding for whether the preferences were competitive or complementary, the number of apples (i.e. 2 or 3), Red’s first choice f_1 , Red’s second choice f_2 and an interaction between f_1 and f_2 . As we fitted the models using half-split cross-validation, it prevented adding further interactions in the model as it led to singular design matrices.

Experiment

Methods

Participants We recruited 100 US participants on Prolific.co (53 male, 46 female, 1 non-binary), aged 42 ± 13 years old. Participants were paid a flat rate of \$12 per hour and the study lasted 14 ± 3.76 minutes.

Design The study was a $2 \times 2 \times 4$ within subject design. First, we varied whether Red and Green had competitive, that is, both prefer apples, or complementary preferences, that is, Red prefers oranges while Green prefers apples. Second, whether the scenes where only Red got to choose had only one apple and one orange or two apples and one orange. Finally, we varied the choices that Red made when Purple was present and when they were absent, yielding four combinations: apple, apple; apple, orange; orange, apple and orange, orange. Overall, combining these conditions yielded 16 stories. The experiment was built using JsPsych (de Leeuw, 2015). The study was preregistered on OSF. The preregistration along with the

data and analysis scripts are available at https://anonymous.4open.science/r/cogsci2026_presentational_goals-5A14

Procedure The study begins with participants seeing instructions about the experiment. Participants are presented with three stories for which they have to answer basic comprehension questions such as: what does Red/Green prefer? Does Purple know what Red/Green prefers? What did Red pick in scene 2 or scene 3? Participants have to answer all of those questions correctly in order to proceed. The main section of the experiment involves two blocks presented in a random order. The first block consists of every story in which the preferences of Red and Green are in competition, that is both Red and Green prefer the apple. We then present, in a random order, every combination of choices Red can make for scene 2 and scene 3. For example, apple when Purple is absent then apple when Purple is present, or apple when Purple is absent then orange when Purple is present. As each set of choices can be applied to stories with one apple and one orange or stories with two apples and one orange, this block is 8 stories long. Following the first block, we let participants know that in the next set of stories, Red will prefer Oranges while Green will prefer Apples. We then repeat the same set of 8 possible combinations of choices and available fruits. After seeing all 16 stories, the experiment ends with a short demographics questionnaire.

After each story, participants were asked three questions, answered on continuous sliders, which we propose map onto

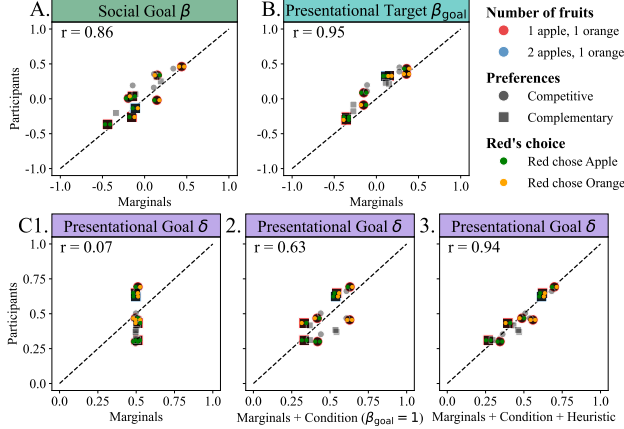


Figure 3: **Correlations between participants’ judgments and model marginals from the joint posterior.** (A) social goal β (how does Red feel about Green?), (B) presentational target β_{goal} (what does Red want Purple to think?), and (C) presentational goal δ (how much does Red care about what Purple thinks of Red?). Panels C2 and C3 show predictions for δ from the model variants that additionally condition on $\beta_{\text{goal}} = 1$ and also add the heuristic.

three key parameters of the model (see Figure 1). We counterbalanced block order, competitive or complementary preferences first, the order of the stories within each block, and the order of scenes 2 and 3 in each story.

Results and Discussion

Fitting procedure The feature-based model was fitted using the MixedLM method of the statsmodel Python package. All other variants were built in Memo (Chandra et al., 2025) and fitted with the Optax Python package (Bradbury et al., 2018). We used gradient descent from random parameter initialisation using a cross-entropy loss, that is we jointly minimised the Kullback-Leibler (KL) divergence between participants’ responses and the marginal posteriors. We cross-validated each model using 100 folds of half-splits: we fitted the model to 8 randomly selected stories and predicting the other 8 for 100 sets of 8 stories.

Systematic intuitions about presentational goals Figure 2 shows participants’ judgments along with predictions from the model using both marginals and the heuristic for δ . We selected 8 stories highlighting the richness and diversity of participants’ inferences. In stories B and F, Red’s social goals are relatively clear as they consistently act opposite to their personal preference in the latter two scenes. Story A illustrates a canonical case of Red yielding their preference to Green in the third scene, when Purple is present, but not in the second scene. Both participants and our model predict that Red has a strong, positive presentational goal (wants to be well perceived by Purple), but both have more uncertainty about Red’s social goal. However, note that our model still considers that the story gives slightly more weight to Red liking Green than

Table 1: **Model comparison.** Median and [5th, 95th] percentiles of correlations (higher is better) and KL divergences (lower is better) for 100 folds of split half cross-validation for each model. The non-feature based models differ only in how they estimate δ .

Pearson’s $r \uparrow$	β	δ	β_{goal}
Feature-based	0.93 [0.60, 0.98]	0.97 [0.87, 0.99]	0.93 [0.66, 0.99]
Marginals	0.85 [0.29, 0.96]	0.08 [-0.39, 0.60]	0.94 [0.17, 0.98]
Marginals + Cond. ($\beta_{\text{goal}} = 1$)	0.85 [0.66, 0.95]	0.61 [0.29, 0.83]	0.95 [0.85, 0.98]
Marginals + Cond. + Heu.	0.86 [0.69, 0.95]	0.96 [0.85, 0.99]	0.94 [0.86, 0.98]
KL divergence ¹ $\downarrow$			
Feature-based	1.23 [1.03, 1.46]	1.52 [1.37, 1.69]	1.14 [0.94, 1.31]
Marginals	0.85 [0.70, 1.01]	1.00 [0.91, 1.08]	0.83 [0.72, 1.01]
Marginals + Cond. ($\beta_{\text{goal}} = 1$)	0.82 [0.70, 1.03]	0.91 [0.79, 1.09]	0.81 [0.70, 0.98]
Marginals + Cond. + Heu.	0.83 [0.69, 1.03]	0.64 [0.56, 0.77]	0.81 [0.71, 0.97]

¹Standardised KL divergence with the uniform distribution as reference: lower than 1 is closer to participants than the uniform, higher is further.

disliking them. Indeed, even if Red has a clear presentational goal, it is not enough to explain away the possibility that they do like Green: after all, they chose to pick an orange when they prefer apples. Participants seem slightly harsher than our model in this case.

The flexibility of presentational targets It is also possible to have negatively valenced presentational goals, such as wanting to hide a friendship from others or be perceived as adversarial. Our model lets agents have presentational targets over any value of β , including when β_{goal} is negative. For example, in story D, Red lets Green take the apple when Purple is absent, but then acts selfishly when Purple is present. Like participants, our model infers that Red truly likes Green, but wants Purple to think that they do not. This may be akin to two childhood friends where one of them is not well liked at school, leading the other to pretend like they are not close in front of other children. Notice the bimodality in the distribution over δ , which may stem from different interpretations of presentational goals. One interpretation ties presentational goals to wanting to be well seen by Purple, which should be low here given that $\beta_{\text{goal}} < 1$. An alternative interpretation is that Red does not actually care what Purple thinks given that their behaviour changed in Purple’s presence. Our model is able to account for both interpretations.

Comparison with feature-based model We compare cross-validated models using Pearson’s correlation between mean judgments and predictions, as well as KL divergence to measure how well models can capture more complex features of participants’ responses (see Table 1). Overall, we see that while the feature-based model captures the mean parameters for each story for each judgments, our model does so

Table 2: **Parameter fits** for each model variant.

Model	τ_{obs}	τ	θ	σ_{goal}	T^{-1}	α	σ_{heu}
Marginals	22.10	0.60	0.89	0.33	3.82		
Marginals + Cond. ($\beta_{\text{goal}} = 1$)	68.93	1.13	0.87	0.36	2.36		
Marginals + Cond. + Heu.	20.22	1.01	0.90	0.36	2.56	0.05	0.04

just as well with fewer parameters. However, KL divergences between the feature model predictions and participants judgments are strictly worse than a uniform distribution. This is because the feature model can only predict tight unimodal distributions around the mean prediction. However, as Figure 2 shows, participants responses are more nuanced. In comparison, our model is strictly better than a uniform distribution at capturing patterns of responses.

Comparison between model variants We now consider variants of the computational model. Table 2 shows parameter estimates. We see that τ_{obs} was consistently high, suggesting participants did not question Red’s true preferences. They however, let Red’s uncertainty about Green and Purple be higher. Furthermore, $\theta \approx .9$ suggests that participants treated Red’s choice in stories with two apples as if they expected Green to be happier we two rather than a single apple. Finally, a σ_{goal} around .33 suggest a relatively tight presentational target around β_{goal} value, suggesting that participants considered Red to have, in general, a clear presentational target. We see that simply using marginal distributions from the joint posterior is sufficient to capture β ($r = .86$) and β_{goal} ($r = .95$). One thing to note with the worse fit to β is that participants tended to answer the midpoint of the range when they were unsure, which is not an issue for the feature-model but a pattern that our model cannot produce: when it is uncertain, its estimate is closer to a uniform distribution.

Figure 3 shows that recovering δ , that is judgments about how much Red cared about what Purple thinks of Red, proved harder. Both conditioning on $\beta_{\text{goal}} = 1$ and the heuristic that a change in Red’s behaviour meant that they care about what Purple thought were needed to capture participants’ responses. In the final full model, the weight α given to the heuristic ended up being 0.05 and its standard deviation $\sigma_{\text{heuristic}}$ was .04. These parameters allowed the model to capture a systematic bimodality in the responses of participants (see Figure 2 C3.). While we recognise that this adjustment is ad hoc, we believe it highlights how hard it can be to make sense of ambiguous social interactions. Notice how for the base model marginalising out other parameters, the KL divergence between participants’ responses and the models’ estimate of δ matches a uniform distribution (Table 1. This is because the model struggles to find a coherent single value for δ given only two scenes. When evidence is sparse, the joint posterior can cohere with several interpretations. It becomes then meaningful to reason over the possibilities using meaningful heuristics or assumptions.

Our inverse planning model matches the performance of three separate feature-based models by jointly estimating Red’s social goal β , presentational goal δ and presentational target β_{goal} . Crucially, it also outperforms it in capturing structural properties of the distributions of participants’ judgments. We find strong evidence that people use inverse planning when it yields clear estimates but also use targeted heuristics to clarify ambiguous situations. This reinforces the propositions that people use inverse planning to reason about ambiguous social

interactions but also for a formulation of presentational goals as information driven utility over how prosocial others perceive us to be.

General Discussion

The present work introduced a novel paradigm and computational model to study how people infer the presentational goals of others. Across 16 different stories, we demonstrate that people can systematically and jointly infer an agent’s social goal (how much they like or dislike another agent), presentational goal (how much they value what another agent thinks), and presentational target (what they want another agent to think about them). Our approach captures a wide range of social interactions where these goals may be misaligned, such as when an agent dislikes another but wants to appear prosocial (Figure 2C), or when an agent likes another but wants to hide this fact (Figure 2D). With minimal assumptions, our model flexibly adapts to each of those situations and matches the participants’ response distributions. This provides additional support for the inverse planning approach as a meaningful representation of how people navigate the social world (Baker et al., 2009, 2017; Jara-Ettinger et al., 2016).

Furthermore, while positively valenced presentational goals may be the norm—we generally want others to think we are *prosocial*—the cognitive process which allows us to infer this goal is not limited to these cases. In fact, both participants and our model readily infer negative presentational goals. As such, we provide evidence that managing impressions rests on a valence-agnostic engine which flexibly adapts to less common social interactions. Our model is intentionally valence neutral; it captures people’s inferences without how these may be used to make higher order judgments. People are remarkably adept at inferring social structures and hierarchies (Davis et al., 2026; Gershman & Cikara, 2020). Formal accounts of the intuitive theories that underlie these inferences are still nascent. Presentational goals may offer a window into this problem. If Red cares what Purple thinks but not what Green thinks, this provides valuable clues about their status, affiliation, or power.

Finally, our findings speak to the computational architecture of theory of mind and what human social utilities are *made of*. First, formalising presentational goals as information-theoretic utilities reveals that what humans consider to be costs and rewards in social interactions goes beyond material outcomes. Instead, they can rely on computations over changes in others’ beliefs which may matter more than material payoffs. Second, while classic false-belief tasks, such as the Sally-Anne task (Wimmer & Perner, 1983), test first-order theory of mind, presentational goals require representing what others believe about my second order utilities: utilities computed over the first-order utility of other people. The fact that participants were systematic in their inferences and well-captured by a rational Bayesian model suggests that human social cognition generally handles this recursion, albeit supported in place by targeted heuristics.

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